

A breakdown of the architecture diagram:

Users

- **Doctors, Admin Staff, Insurance Teams** - These are the primary end-users who interact with the system.

Frontend Layer

- **Streamlit App** - Provides the user interface for chat, uploads, dashboards, and interactions.
- **FastAPI Gateway** - Acts as the entry point, routing requests from the frontend into the orchestration layer.

Central Orchestration (LangGraph Supervisor)

This is the “brain” coordinating multiple specialized agents:

- **Planner Agent** - Breaks down tasks into subtasks.
- **Retriever Agent** - Performs vector search and document ranking.
- **MCP Agent** - Executes tools and queries the PostgreSQL database.
- **Vision Agent** - Handles image analysis, OCR, and extraction.

LLM Processing & Reasoning

- **LLM Engines (OpenAI, Groq)** - Provide language understanding and generation.
- **Reasoning Agent** - Synthesizes context, applies logic, and inference.
- **Evaluator Agent** (LLM-as-a-judge)- Ensures quality, safety, and validation of outputs.
- **Response Agent** - Applies guardrails before delivering the final answer.

Data Ingestion & Storage

- **Ingestion Pipeline** - Handles document upload, chunking, and embedding creation.
- **Vector Database (FAISS / ChromaDB)** - Stores embeddings for semantic search.
- **PostgreSQL Database** - Contains structured hospital data (patients, appointments, labs, billing).
- **MCP Server** - Provides secure access to databases and tools.

Observability & Monitoring

- **LangSmith** - Tracks performance, logs, and observability for debugging and evaluation.

Workflow :

1. **Users** interact via the Streamlit app.
2. Requests go through **FastAPI Gateway** → **LangGraph Supervisor**.
3. Supervisor delegates tasks to specialized **Agents** (Planner, Retriever, MCP, Vision).
4. Agents query **Vector DB** or **PostgreSQL** via **MCP Server**.
5. Results are processed by **LLMs**, refined by **Reasoning/Evaluator Agents**, and finalized by the **Response Agent**.
6. **Observability tools** monitor the entire pipeline for safety and performance.